

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Regular Summer 2024 Course: B. Tech. Branch : Electronics and Computer Engineering Semester :VI Subject Code & Name: BTECOE604A VLSI Design Max Marks: 60 Date:21/06/2024 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Explain Ids and Vds characteristics of depletion and enhancement type mosfet.	CO1	6
B)	Explain CMOS Inverter analysis and design. Implement a three-input NOR gate using CMOS logic.	CO4	6
C)	Explain pass transistor logic with neat diagram.Implement NAND and NOR gate using PTL.	CO2	6
Q.2	Solve Any Two of the following.		12
A)	Design stick diagram for: i) $F = \overline{(A.B)+C}$ ii) $F = \overline{A+B+C}$	CO2	6
B)	What is Stick Diagram? What are the uses of Stick diagram.Draw the stick diagram and layout for CMOS inverter.	CO2	6
C)	Differentiate between NMOS,CMOS and Bi CMOS inverters.	CO1	6
Q. 3	Solve Any Two of the following.		12
A)	Explain transmission gate with its symbol.Implement NOR gate using TG.	CO2	6
B)	Explain Pseudo nMOS Logic, Dynamic CMOS LOGIC and clocked CMOS logic.	CO1	6
C)	Explain different alternate gate circuits.	CO2	6
Q.4	Solve Any Two of the following.		12
A)	Explain twin tub process for CMOS fabrication.	CO1	6
B)	With neat diagram explain different operating regions for a MOS transistor.	CO1	6

C)	Give the subsystem design considerations of a four-bit adder.	CO3/CO4	6
Q. 5	Solve Any Two of the following.		12
A)	Explain about different programmable elements in FPGA architectures.	CO5	6
B)	Implement the following functions using PLA. $A(x,y,z) = \sum m(1,2,4,6)$ $B(x,y,z) = \sum m(0,1,6,7)$ $C(x,y,z) = \sum m(2,6)$	CO4	6
C)	Write about the following i) Transistor-transistor Logic (TTL) ii) Emitter – coupled Logic (ECL) iii) CMOS Logic	CO4	6
	*** End ***		